

Investigation of the bacteria on different surfaces in the school compound

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Abstract

Most schools do not know where the most bacteria-contaminated surface within their compound is. Research has shown that the bacteria commonly found on surfaces can all be extremely harmful. This study investigates the presence of bacterial contamination in various areas of SST. We selected tables, door handles, walls, floors, windows, and canteen seating for sampling. We used swab sticks to collect samples and grew them on nutrient agar plates. After growing the bacteria on the agar for about a day and a half, the plates were taken out and examined. Analysis of the plates demonstrated that the canteen table had the highest number of colonies, while the control surface swabbed immediately after cleaning had none. The research aims to inform janitors and students about the surfaces that need to be cleaned frequently to reduce the spread of infectious diseases in schools. The report concludes that frequently touched surfaces like desks, doorknobs, and computer mice have a higher bacterial count than less regularly touched surfaces like walls and floors.

Keywords: Bacterial contamination, Frequency of touching, colonies

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Project Title: Investigation of the bacteria on different surfaces in the school compound

1. Introduction:

1.1. Background information

The investigation of bacteria on different surfaces in a school compound is an important area of research that pertains to the overall cleanliness and hygiene of the school environment. This research aims to identify sources of bacterial contamination in the school and assess their potential impact on student and staff health. Studying the types of bacteria on different surfaces can help determine where we can improve to reduce contamination and promote good hygiene practices. This investigation addresses the immediate concerns of student and staff health.

1.2. Specific topic

The investigation of bacteria on different surfaces in a school compound is a specific topic of great importance due to the potential impact on student and staff health. Poor hygiene in schools can lead to various health problems, including respiratory and skin infections (Busch-Vishniac, 2003). School environments have a crucial role in the health and development of children; it is essential to ensure that it is safe and hygienic. (Kalia, Virender, and Divya Sood, 2017). This research aims to identify sources of bacterial contamination in schools. Frequently touched surfaces like desks, doorknobs, and computer mice tended to have a higher bacterial count than less touched surfaces like walls (Shimizu, Y, 2020). By identifying the sources of bacterial contamination in schools, it can help to reduce the spread of infectious diseases in schools, promote overall health and well-being, and contribute to the goal of ensuring healthy lives and promoting well-being for all at all ages, under the United Nations' Sustainable Development Goals.

1.3. Past research

Previous research on bacterial contamination in schools has focused on the potential impact of poor hygiene on student and staff health.

Poor indoor air quality and bacterial contamination can cause respiratory issues, allergies, and infections, leading to increased absenteeism and decreased academic performance (Busch-Vishniac, 2003). Proper hygiene in schools is important to reduce the spread of infectious diseases, like currently, SARS-CoV-2 (Hall and Jensen, 2002). One of the most effective ways to reduce the bacterial contamination is education on proper hand hygiene techniques and having hand sanitisers in the classroom (Kalia and Sood, 2017). We do not have hand sanitisers in the classroom, we only have surface cleaners.

1.4. Conclude the Introduction

1.4.1 Specific objectives of the research

In conclusion, this research aims to identify sources of bacterial contamination in a school compound by collecting samples from various surfaces and identifying the types of bacteria present.

1.4.2 Research Questions

Our research question is as follows:

“How does the frequency of a surface being touched affect the amount of bacteria on it?”

1.4.3 Research Hypotheses

We decided to test the following hypothesis: z

H1: “When a surface is touched more frequently, the number of bacteria in number of colonies and the area of the colonies will also increase exponentially.”

1.4.3.1 Independent variable

The independent variable is the areas the bacteria are taken from.

1.4.3.2 Dependent variable

The dependent variable is the number of bacteria colonies and size of bacteria colonies

1.4.3.3 Controlled variables

- (a) The swabbing day should be the same day.
- (b) The growing period should be the same time. (e.g. 24 hours)
- (c) How to get a sample (e.g. dilution)

2. Method

2.1 Equipment list

- 8 x 2 sterile swab sticks with sterile container
- 8 x 2 nutrient agar plates with covers
- 2x set of sterile gloves
- 1x Pen
- 24x sticker labels
- Incubator oven (37°C)

2.2 Diagrams

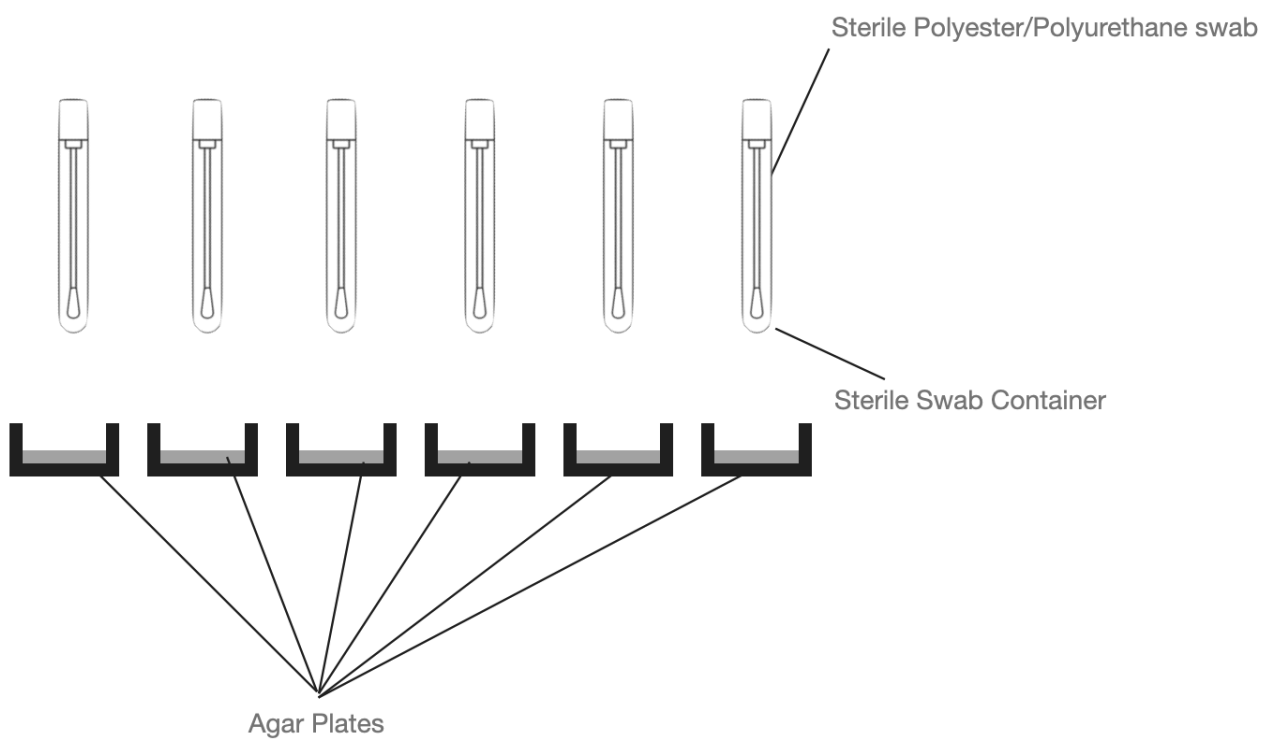
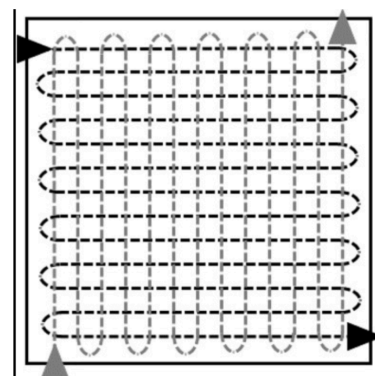


Figure 1: Experimental setup

Detail all procedures and experimental design to be used for data collection

1. Set up the experiment as shown in the diagram above.
2. Identify seven areas you want to swab (4cm by 4cm area)
 - a. ensure that the frequency of these surfaces being touched varies.
 - b. suggestions include
 - i. table surface
 - ii. door handles
 - iii. wall
 - iv. door itself
 - v. floor
 - vi. table that isn't cleaned frequently
 - vii. a seat
3. Swab the first area following the pattern in Figure 1.1
 - a. Sanitize your hands before and after swabbing.
 - b. Follow the black arrow, followed by the grey arrow
 - c. Ensure you do not press on the neck of the swab
4. Place the swab back into the container and label the container where the swab was taken
5. Repeat steps 3 - 4 for the next 6 areas and on a surface cleaned in the last minute. (label the cleaned one as "Control")



6. Put the agar plate on a sterile table with the containers of the swabs.
7. Gently swipe the control swab onto the agar plate following the pattern in Figure 1.2
 - a. Follow arrow 1, then 2, then 3.
 - b. Rotate the swab while smearing.
8. Cover the agar plate with the cover and label it as “control”
9. Repeat steps 6 - 7 for the other 7 swabs. Note to label the cover of the agar plate to the corresponding label of the swab location.
10. Put all agar plates in an incubator oven at 37°C for 3 days

2.3.2: Analysing bacteria

11. Take the agar plates out of the incubator and put on gloves.
12. Take a photo of the agar plate from the top, with a reference like a ruler on the side
13. Upload the photo somewhere
14. Count the number of colonies
15. Use the formula area of a circle = πr^2 to calculate the area of the amount of bacteria.
 - a. We used $\pi = 3.14$ to calculate
 - b. Use a program to calculate to make it easier
 - i. Suggestions:
 1. Tracker
 2. Keynote
16. Repeat steps 12 - 15 for the other plates
17. Repeat the experiment with the same surfaces and average the results for increased accuracy

2.3.3: Collating data

2.3.3.1: Tables

1. Collate the data in the following table:

Surfaces swabbed (In order of frequency of touching)	Reading 1		Reading 2	
	Number of colonies	Area of colonies (2d.p)	Number of colonies	Area of colonies (2d.p)
Control	1	0	1	0
Surface 1	2	0.1	2	0.1
Surface 2	4	0.2	4	0.2
Surface 3	8	0.4	8	0.4
Surface 4	16	0.8	16	0.8
Surface 5	32	1	32	1
Surface 6	64	3	64	3
Surface 7	128	6	128	6

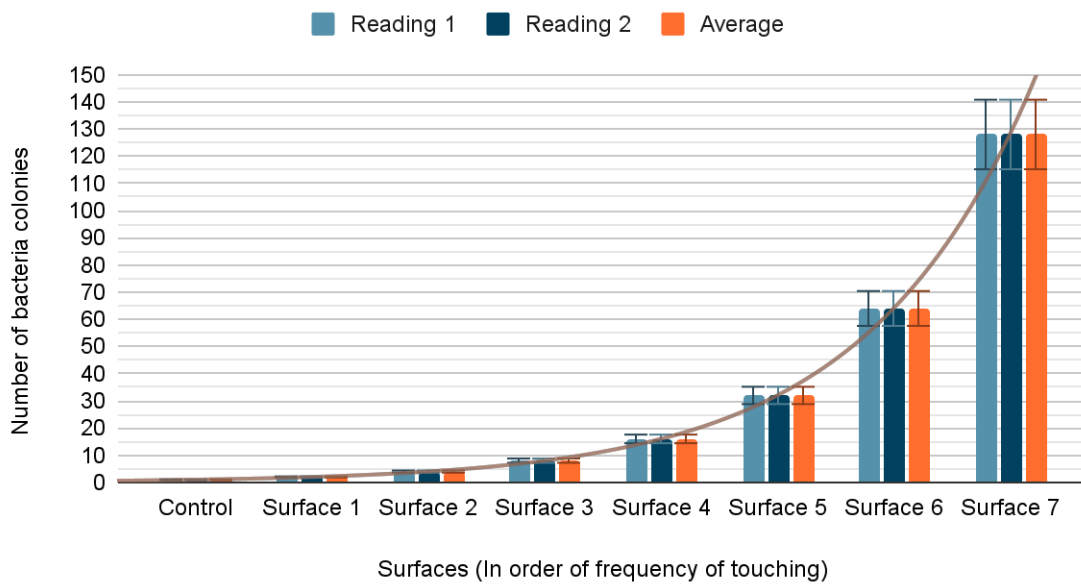
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Surfaces swabbed (In order of frequency of touching)	Average		Stn Dev.	
	Number of colonies	Area of colonies (2d.p)	Number of colonies (Nearest whole number)	Area of colonies (2d.p)
Control	1	0	0	0
Surface 1	2	0.1	0	0
Surface 2	4	0.2	0	0
Surface 3	8	0.4	0	0
Surface 4	16	0.8	0	0
Surface 5	32	1	0	0
Surface 6	64	2	0	0
Surface 7	128	4	0	0

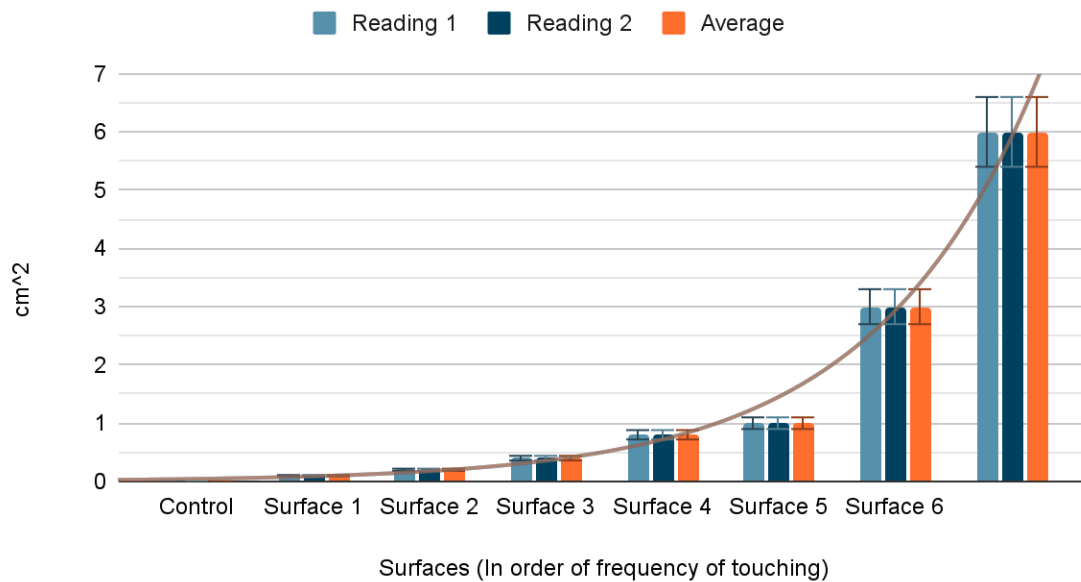
2.3.3.2: Graphs

2. Draw the graphs like so:

Number of Colonies



Area of Bacteria (2 d.p)



2.3.3.3: Reading the Graphs

Based on the trend of the graph, we can tell if the frequency of touching affects the amount of bacteria on the surfaces exponentially. In this case, it does.

If the trendline goes down all of a sudden, either the surface is not touched often, or the frequency of touching does not affect it much.

If the trendline goes down continuously, the frequency of touching reduces the amount of bacteria on a surface

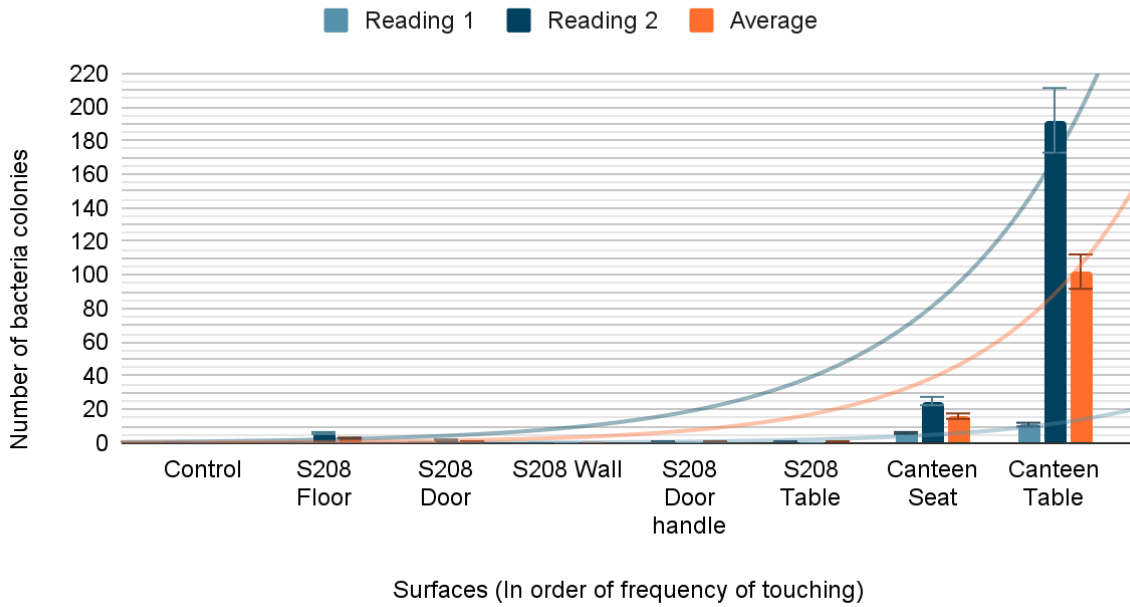
3. Results

3.1: Tables

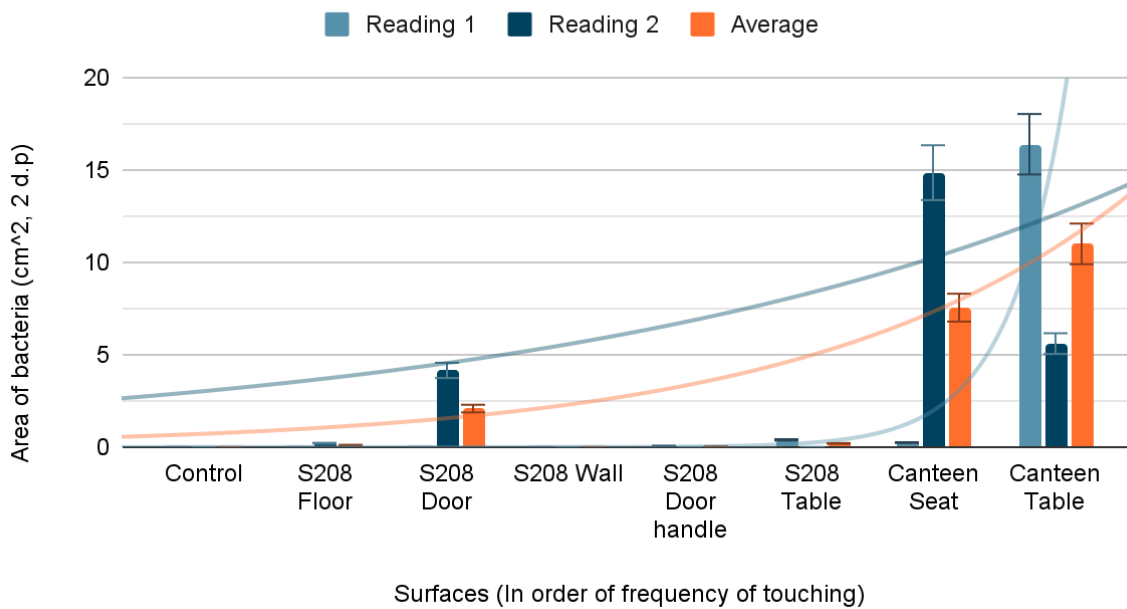
Surfaces swabbed (In order of frequency of touching)	Reading 1		Reading 2	
	Number of colonies	Area of colonies (2d.p)	Number of colonies	Area of colonies (2d.p)
Control	0	0.00	0	0.00
S208 Floor	0	0.00	6	0.21
S208 Door	0	0.00	2	4.15
S208 Wall	0	0.00	0	0.00
S208 Door handle	1	0.06	0	0
S208 Table	1	0.39	0	0
Canteen Seat	6	0.24	25	14.84
Canteen Table	11	16.38	192	5.60
Surfaces swabbed (In order of frequency of touching)	Average		Stn Dev.	
	Number of colonies	Area of colonies (2d.p)	Number of colonies	Area of colonies (2d.p)
Control	0	0.00	0	0
S208 Floor	3	0.11	3	0.11
S208 Door	1	2.08	1	2.08
S208 Wall	0	0.00	0	0
S208 Door handle	1	0.03	1	0.02
S208 Table	1	0.20	1	0.10
Canteen Seat	16	7.54	10	7.30
Canteen Table	102	10.99	90	4.52

3.2: Graphs

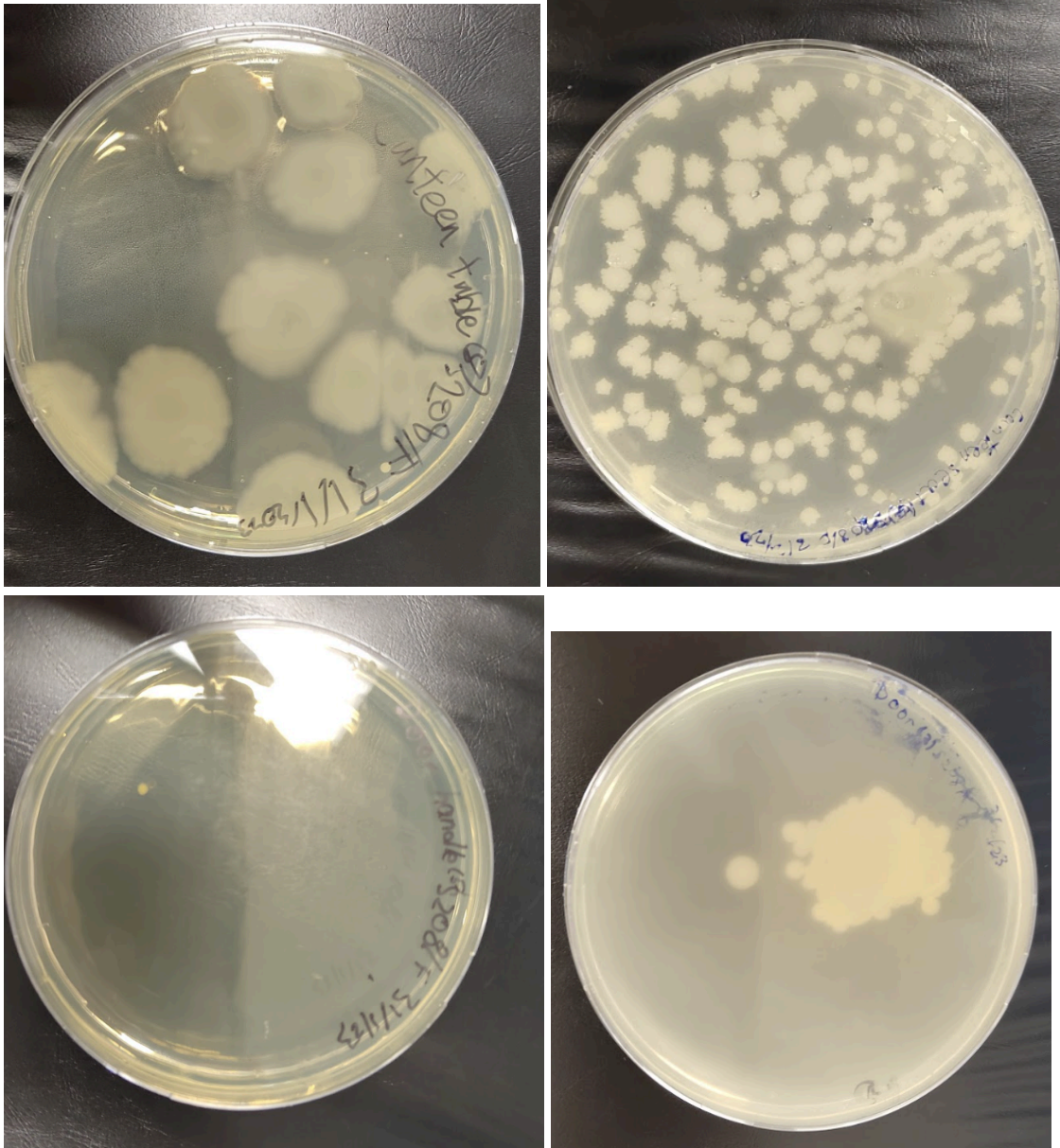
Number of colonies



Area of bacteria (2 d.p)



3.3: Photos



The first photo is of canteen table bacteria. It holds the most bacteria colonies and biggest bacteria size. Second photo is the canteen seat. It holds the second most bacteria colonies and second biggest bacteria size. Third photo is the door handle. It was expected to have more colonies and bigger bacteria size but fails to meet the hypothesised amount greatly. Fourth photo is the door. It was expected to have more colonies but it was close. The bacteria size is also similar to the hypothesised one.

4. Discussion

4.1: Key Findings, Comparison and explanations

The canteen table had the most and biggest size colonies. The control table we had cleaned before swabbing had none during the whole experiment, as we hypothesised. We listed down eight sites to swab for bacteria. We had not expected the door handle to be so clean. We had also thought the door handles would be a highly touched surface.

The canteen tables have the most colonies with 203 colonies in total, followed by the canteen seat with 48 colonies in total, then the S208 floor with 6 colonies in total, then the S208 door with 2 colonies in total, the S208 door handle and S208 table that is not cleaned with a colony each, with the S208 wall and control at last with 0 colonies each.

The canteen tables have the biggest size colonies with 21.98 cm^2 , followed by canteen seat with 15.08 cm^2 , S208 door with 4.15 cm^2 , S208 table with 0.39 cm^2 , S208 floor with 0.21 cm^2 , S208 door handle with 0.06 cm^2 , S208 wall and control with 0.00 cm^2 .

For bacteria colonies, it can be observed that 5 of our predicted frequencies (left to right, lowest to highest) are accurate, the three exceptions being the S208 door handle, S208 door, and S208 table. All three had fewer colonies than hypothesised. It might be possible that the S208 door handle, S208 door, and S208 table are not as frequently touched as thought so. From most of the results, we can conclude that the more touched a surface is, the more bacteria there will be on it.

The trendline of our experiment also supports the theory that the amount of bacteria on a surface will multiply faster and faster with each touch. This is known as the exponential growth of microorganisms, where the number of microorganisms doubles with each replication cycle. Thus, a surface that is frequently touched will have a higher number of colonies compared to a surface that is rarely touched.

Additionally, the unexpected finding that the door handle, door, and table had fewer colonies than hypothesized can be explained by the theory of hygiene. This theory states that the level of bacterial contamination on a surface is influenced by several factors, including the frequency of cleaning and the level of contact with the surface.

Hence, it is possible that these surfaces were cleaned more frequently than anticipated, or they were not touched as often as expected. Overall, the key findings of our experiment can be explained by the principles of microbial ecology and hygiene, which are essential in understanding the distribution and growth of microorganisms on surfaces.

More frequently touched surfaces have higher levels of bacterial contamination (Shimizu, Y, 2020). This research supports this and gives schools an idea of the trend and relationship between the frequency of touching and the number of bacteria on it.

4.2: Limitations and Recommendations

I believe we could have collected more data samples if we were not limited to the school grounds. We would have more accurate and reliable results if we took samples from public places. But to avoid complicating the experiment, we only took samples from the school. Additionally, we wanted to get more relevant data for the school.

Another limitation would be the inability to perform a gram stain on the bacteria collected. As the bacteria was unknown, we were unable to perform a gram stain. This prevented us from truly understanding the bacteria on the surfaces and whether they threatened the students' and school staff's well-being and health.

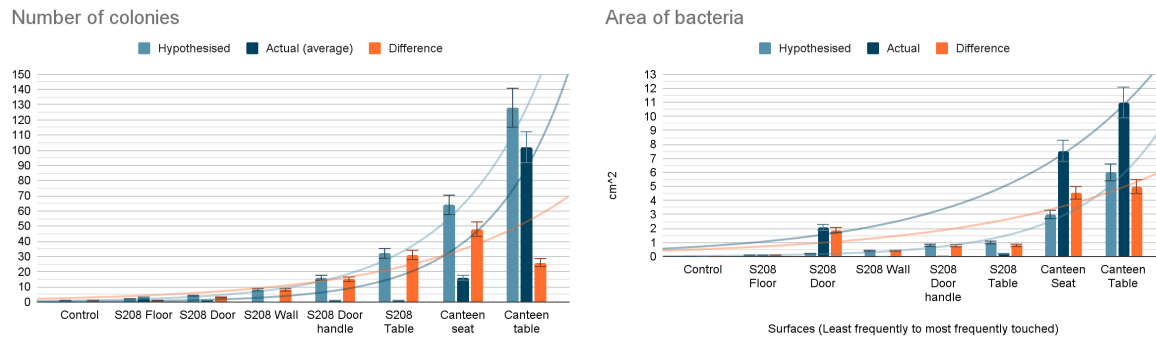
Another limitation was the time. Due to the deadline of the data collection, we could not grow the bacteria for very long and hence the cleaner objects like the control and the wall would be portrayed as equally clean, but this could not be the case.

A possible recommendation would be to take data from several schools. For example, the surfaces from New Town Secondary School. This would allow the data to be less localised to just SST and appeal to a more general audience of different schools in Singapore. By making the data recorded more relevant to other schools, the data could be used by different schools to step up their sanitation techniques, making the research even more relevant to good health and well-being, one of the seventeen UN sustainable development goals.

A possible recommendation is to perform a gram stain. This would require a Personal Protective Equipment as the bacteria on the surfaces are unknown and hence it is not allowed to do the gram stain. The bacteria, as it is unknown, will require high levels of security to perform the stain.

Another recommendation would be to grow the bacteria for longer. In our experiment, we only grew each batch of bacteria for 3 days due to time constraints. Should we grow the bacteria slightly longer (e.g. 1 week), the experiment would be more accurate, as you can accurately compare how much bacteria is on each surface. In our dataset, we are unable to compare the bacteria from the wall and the control, as both show up as having no bacteria on them at all, while the wall should have more bacteria than the control, as the control surface was cleaned right before swabbing.

4.3: Evaluation of Hypothesis



I believe that our hypothesis is partially accepted. Highly touched surfaces like canteen tables and seats do have the most and biggest size colonies.

We had expected the door and door handle to have more, but it makes sense that they do not have much bacteria. In SST, the doors are generally left open during and outside class. We also expected that the floor would have bacteria as most students would not clean the soles of their shoes often, and it is not uncommon for people to step in food or other bacteria-high places.

Overall, I think the results do not differ too much from our hypothesis.

5. Conclusions

5.1: Summary of findings

The purpose of this study was to identify sources of bacterial contamination in a school environment and determine the relationship between the frequency of touching a surface and the amount of bacteria present on it. The hypothesis was that the frequency of touching a surface would have an effect on the amount of bacteria present on it. The research was conducted by collecting samples from eight different areas in a school environment, including surfaces such as tables, door handles, walls, floors, windows, and canteen seating, and culturing them on nutrient agar plates. The results indicated that the canteen table had the highest number of colonies, while the control surface cleaned regularly had none. The door handle had minimal colonies, and the door was surprisingly clean. The hypothesis was mostly verified, as the higher levels of bacteria on the canteen table compared to other surfaces did suggest that the frequency of touching a surface does have an effect on the amount of bacteria present.

5.2: Contributions of research

Our research has two contributions:

Firstly, it identifies the areas with the most bacterial contamination in a school environment. It provides information on where cleaning efforts should be focused to reduce the risk of spreading infectious diseases.

Secondly, it explores the relationship between the frequency of touching a surface and the number of bacteria on it, which can improve the school's hygiene and sanitation practices.

Overall, the research provides valuable insights into the bacterial contamination of SST, and its findings can improve students' and staff's health and well-being, contributing to the UN local and global goal of good health and well-being.

5.3: Practical Applications

The results of this study can be used by the janitors to implement strategies to reduce the presence of harmful bacteria on frequently touched surfaces. For example, we could increase the frequency of cleaning high-touch areas such as the canteen table, or implement hand hygiene programs for students and staff. A reference to other similar studies can be made, such as "The effectiveness of hand hygiene interventions in reducing absenteeism in schools" (Curtis, 2019).

5.4: Areas for further study

Further research could be conducted to investigate the efficacy of various cleaning methods for reducing the presence of bacteria on frequently touched surfaces in a school environment. This could include an examination of the effectiveness of different cleaning agents, as well as the frequency of cleaning needed to maintain a low level of bacteria.

The independent variable would be the cleaning and disinfection method used, while the dependent variable would be the level of bacterial contamination on the surfaces after cleaning.

Certain cleaning and disinfection methods were more effective than others in reducing bacterial contamination on surfaces in healthcare settings (Rutala and Weber, 2013). However, there needs to be more research on the effectiveness of these methods in school settings.

Therefore, further research in this area could help identify effective cleaning and disinfection strategies for reducing bacterial contamination on school surfaces, which could help reduce the spread of infectious diseases among students and staff.

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